

Jenvith M Machine Learning Engineer (617) 752-6363 jenvithm10@gmail.com Boston, MA LinkedIn: https://www.linkedin.com/in/jenvithm/	
SUMMARY	Machine Learning Engineer with 3+ years of experience designing and deploying production-grade ML and LLM systems across domains. Proven track record of building RAG-based multi-agentic AI frameworks and high-throughput inference pipelines on cloud-native platforms (AWS, Azure, GCP). Skilled in optimizing LLMs and foundational models with LoRA, fine-tuning, quantization, and memory-efficient serving for Natural Language Processing (NLP) applications, reducing latency and cost at scale. Expert in MLOps/LLMOps, CI/CD automation, and governance using Kubeflow, SageMaker, Terraform, and GitOps, ensuring 99.99% uptime and fully reproducible experiments. Passionate about delivering scalable, reliable, and cost-efficient AI solutions that enhance business outcomes.
SKILLS	Programming: Python, SQL, PySpark, Java, Scala, R, Bash ML/DL: PyTorch, TensorFlow, XGBoost, LightGBM, CNN, RNN, LSTM, Transformers LLMs & GenAI: LoRA, QLoRA, Quantization, LangChain, LangGraph, LangSmith, vLLM, Hugging Face, OpenAI API, Agentic Workflows, RAG, Vector Search (FAISS, Pinecone), Ollama, Machine Learning Lifecycle MLOps / LLMOps: Kubeflow, MLflow, SageMaker Pipelines, Docker, Kubernetes (EKS), KServe, TorchServe, ArgoCD, FastAPI, Feast (Feature Store), vertex AI, CI/CD, REST APIs, GitHub Actions, Terraform, Helm, Jenkins, AIOps, GPU Optimization Monitoring & Governance: SHAP, LIME, Evidently AI, Prometheus, Grafana, Great Expectations, Drift Detection, Model SLOs Cloud & Data Platforms: AWS (SageMaker, S3, Lambda, EKS, Redshift, EMR, EC2), GCP, Azure ML, Apache Spark, Kafka, Flink, Snowflake, BigQuery, Hive, distributed systems, Dbt cloud Databases & Visualization Tools: PostgreSQL, Redis, MongoDB, Elasticsearch, Cassandra, Tableau, Power BI, Streamlit, Plotly Compliance: Data Governance, GDPR, SOX, Responsible AI Practices
WORK EXPERIENCE	Piper Sandler, USA Machine Learning Engineer Sep 2024 – Present - Architected and launched a multi-agent AI RAG framework with LangGraph, LangChain, and vector databases (Pinecone/FAISS), incorporating AI Agents with advanced prompt engineering techniques for more accurate and explainable outputs. - Enhanced AI agent workflows with clearer role distinction of AI agents and memory management, leveraging embeddings and extended context windows for higher accuracy. - Built specialized analyst agents for sentiment analysis, fundamental research, and technical/macro signals by leveraging embeddings and extended context window management, cutting manual research workload by 40%. - Deployed LoRA fine-tuned LLaMA-3.1-7B foundational models with vLLM and integrated MCP servers for secure, scalable enterprise inference, delivering cost-efficient, production-ready deployments. - Optimized LLM inference at scale by implementing KV caching, quantization, and memory-efficient techniques, and GPU optimization strategies, resulting in 35% lower latency and 40% reduced memory usage, boosting throughput and user experience. - Designed and deployed production-grade ML models on AWS EKS using Kubeflow, KServe, and ArgoCD, with inference pipelines handling 50K+ predictions/minute and 99.99% uptime. - Implemented zero-downtime rollouts, reducing model deployment time by 60% and improving reliability for production workloads. - Standardized CI/CD and MLOps pipelines using Terraform and GitOps, improving release velocity by 45% across 10+ ML models. - Ensured audit-ready logging, data lineage, and governance with Kubeflow Pipelines and MLflow, achieving 100% reproducible experiments and zero failed releases. Quincy Credit Union, USA Machine Learning Engineer January 2024 – July 2024 - Architected an end-to-end ML pipeline for a real-time fraud detection system, leveraging Docker and Jenkins to create a CI/CD pipeline that reduced manual deployment time by over 90%. - Designed and built comprehensive data pipelines for the fraud system, including batch ETL jobs for model training and a real-time Apache Spark Streaming pipeline for inference, constructing features with sub-second latency and versioning data with DVC. - For the fraud system, developed and tuned Isolation Forest and LSTM models in MLflow to address severe class imbalance; the final model improved fraud detection recall by 15% over the previous baseline. - Deployed the hybrid fraud model on the Kubernetes (AWS EKS), creating an automated retraining and deployment pipeline with Apache Airflow that achieved Less than 100ms latency and reduced infrastructure costs by 40%. - Established a monitoring stack for the fraud model using Prometheus/Grafana and an SNS-to-SQS alerting system that reduced critical incident detection time to under 60 seconds. - For a separate loan default prediction task, built a scalable batch ETL pipeline to process and aggregate raw transactional data, engineering over 30 features using PCA and mutual information to boost F1-score by 12%; the resulting stacked ensemble model (XGBoost, Logistic Regression) achieved a 0.92 AUC. Cyient, India ML Engineer January 2020 - July 2022 - Designed and Productionized ML models using AWS SageMaker and AWS EKS, enabling streaming inference pipelines that reduced false positives by 20% and improved detection accuracy by 15%; implemented automated retraining and model rollouts via Kubeflow, GitHub Actions, and Jenkins. - Built a scalable feature store with AWS Glue and EMR (PySpark) to deliver fresh, versioned features for both batch and streaming models, improving model freshness and reproducibility across environments. - Implemented ML model monitoring and explainability by integrating SHAP outputs, drift detection using KS test, and performance metrics into CloudWatch and Prometheus/Grafana, ensuring reliable production performance and proactive retraining triggers. - Designed and deployed a real-time fraud detection data pipeline using Apache Kafka and Spark Structured Streaming, processing 500K+ transactions daily with sub-second latency, integrated with SNS for instant fraud alerts and SQS for reliable queue management. - Developed automated ETL workflows and analytics pipelines using Airflow and AWS Glue, managing 1TB+/month of transaction data stored in S3 and curated into Amazon Redshift, while enforcing compliance via AWS KMS encryption, IAM roles, and CloudTrail audit logging.
EDUCATION	Masters in Data Science Northeastern University, Boston, MA, USA May 2024 Bachelors in Technology SRM University, India May 2021